Candidate Seat No__________

# CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

**Fifth Semester of B. Tech. (IT) Examination**

**November 2022**

**IT351 Design & Analysis of Algorithms**

**Date: 26.11.2022, Saturday** **Time: 10:00 a.m. To 1:00 p.m.** **Maximum Marks: 70**

***Instructions:***
1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.

## SECTION – I

**Q - 1** **Do as directed.**

**1)** What is growth rate of an algorithm? **[01]**

**2)** Derive the Big-oh time complexity for the given code. **[01]**

*while (n >=2)*
  $n = \sqrt{n}$

**3)** Write general recurrence relation for the Divide and Conquer strategy. Define each term of the recurrence relation. **[01]**

**4)** Consider the following functions: **[01]**

$f(n) = 2^n$
$g(n) = n!$
$h(n) = n^{\log n}$

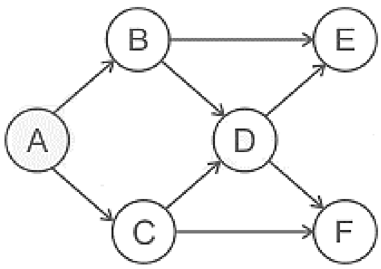
Which of the following statements about the asymptotic behavior of f(n), g(n), and h(n) is true?

(A) f(n) = O(g(n)); g(n) = O(h(n)) (B) f(n) = Ω (g(n)); h(n) = O(h(n))

(C) g(n) = O(f(n)); h(n) = O(f(n)) (D) h(n) = O(f(n)); g(n) = Ω (f(n))

**5)** Consider Merge sort algorithm takes 30 seconds in the worst case for an input of size 64. Calculate maximum input size of a problem that can be solved in 6 minutes. **[01]**

**6)** What is topological ordering of Graph? List out all different topological ordering for the given Graph. **[02]**



**Q-2 (a)** Which design technique is used to solve Quick sort algorithm? Analyze the worst case and average case performance of the Quick sort algorithm with appropriate input scenario and recurrence relation. **[04]**

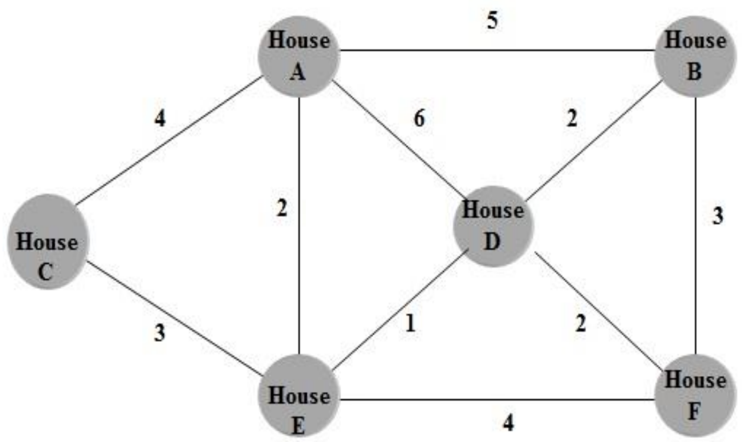
**Q-2 (b)** **Attempt following Questions (Any Two).** **[10]**

1) Find the optimal solution for fractional knapsack problem where the capacity of the knapsack is 28 Kg and there are seven items with the given profit and weight.
profit = {9, 5, 2, 7, 6, 16, 3}, weight = {2, 5, 6, 11, 1, 9, 1}

2) Derive mathematical equation and solve it for worst case time complexity of the given algorithm.

```
Function (A: An Array)
    for j ← 2 to n
        do key ← A[ j ]
            i ← j - 1
            while i > 0 and A[i] > key
                    do A[i + 1] ← A[i]
                        i ← i – 1
            A[i + 1] ← key
```

3) Consider cable television company laying cable to a new neighborhood. A given graph shows that different houses are connected by certain cables, it is required to bury the cables solely along specific paths. Some of those paths might be more expensive, because they are longer, these paths would be represented by edges with larger weights.
   (A) Identify appropriate algorithm and design approach, so that the cable television operator is able to connect every house with minimum cost of cable.
   (B) Construct minimum cost network for the given graph.




**Q-3 (a)** Construct pruned state space tree for 4-Queen problem using Backtracking approach. Show the solution vector with implicit and explicit constrain for 4-Queen problem. **[06]**

**Q-3 (b) Attempt following Questions (Any Two).** **[08]**

**1)** Design the recursive algorithm for the summation of first N natural numbers. Derive the recurrence relation and solve it using backward substitution method.

**2)** Discuss general characteristics of Greedy Approach. Why Greedy Approach does not guarantee to generate global optimal solution to every optimization problem? Justify with an example.

**3)** Apply Branch and Bound strategy to solve least cost Job Assignment problem with the following cost matrix. Consider 4 persons (A, B, C, D) and 4 jobs (J1, J2, J3, J4).

| | **J1** | **J2** | **J3** | **J4** |
|---|---|---|---|---|
| **A** | 9 | 2 | 7 | 8 |
| **B** | 6 | 4 | 3 | 7 |
| **C** | 5 | 8 | 1 | 8 |
| **D** | 7 | 6 | 9 | 4 |

## SECTION – II

**Q - 4 Do as directed.**

**1)** Arrange the given terms in increasing order of their Big-Oh time complexity. **[01]**

*O(n ), O(n^(3/2)), O(log(log n)) , O(n), O(n²log n), O(2ⁿ), O(n!), O(nlogn)*

**2)** Is Dynamic Programming a top-down or bottom-up approach? Justify. **[02]**

**3)** Compare Dynamic Programming with Divide and Conquer approach. **[02]**

**4)** Calculate number of ways to multiply three matrices (P, Q and R) is ____________ and minimum number of multiplications required is ___________, where dimensionality of matrices is $P_{4x2,}$ $Q_{2x4}$ and $R_{4x1}$. **[02]**

**Q-5 (a)** Demonstrate optimal substructure property for Making Change Problem using Dynamic Programming. **[04]**

**Q-5 (b) Attempt following Questions (Any Two).** **[10]**

**1)** Discuss Big-Oh, Omega and Theta notations with appropriate graph.

**2)** What are the limitations of Master Theorem? Solve the following recurrence relations using Master Theorem.

(A) $T(n) = 4T\left(\frac{n}{2}\right) + \frac{n}{logn}$

(B) $T(n) = 16T\left(\frac{n}{4}\right) + n!$

**3)** Prove that recursive solution of Fibonacci series takes running time of $O(2^n)$. How to improve the exponential performance of Fibonacci series?

**Q-6 (a)** Define class P, NP, NP-Hard and NP-Complete problems and show relationship among them. **[04]**

**Q-6 (b)** **Attempt following Questions (Any Two).** **[10]**

**1)** Evaluate Big-Oh time complexity of the following recurrence relation using recurrence tree method.

$$T(n) = T\left(\frac{n}{10}\right) + T\left(\frac{9n}{10}\right) + n$$

**2)** What is time complexity of Rabin-Karp string matching algorithm? How many spurious hits does the Rabin-Karp string matching algorithm encounter in the text T = "41531265392" when looking for all occurrences of the pattern P ="26" with working modulo q="11" and numbers {0,1, 2, ...., 9}? Show all the steps with spurious hits.

**3)** Consider the LCS of two amino-acid sequences H E A G A W G H E E and P A W H E A E. The names of the corresponding amino-acids are A: Alanine, E: Glutamic acid, G: Glycine, H: Histidine, P: Proline, and W: Tryptophan. Find one Longest Common Subsequence of given two strings using Dynamic Programming.

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